Inclusiveness Detection in Bangla Textbooks: Exploring LLMs and Advanced Prompting Techniques

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***Abstract*—In our rapidly growing world, addressing the increasing social, cultural, political, and historical concerns is very important. One of the most essential of these issues is inclusiveness, the representation and involvement of people from all genders, religions, cultures, races, and so on. Despite its importance, inclusiveness is often overlooked, leading to social conflicts and debates, especially in the context of educational materials. In this study, we evaluate the capability of large language models (LLMs) to detect inclusiveness in Bangla textbooks. We evaluated three state-of-the-art LLMs: GPT-4o (OpenAI), Gemini-1.5-Flash (Google AI), and Mistral Large 2.1 (Mistral AI). Moreover, there is no existing study on effectively prompting LLMs for context-aware inclusiveness detection, especially in Bangla textbooks. Therefore, we used these three LLMs with three different prompting techniques: Direct Prompting (DP), Prompting with Definition (PwD), and Chain-of-Thought (CoT) prompting on each LLM. Our findings demonstrate that different models interpret the same data differently, and even within a single model, prompting techniques yield diverse outputs, revealing a methodological gap in evaluating inclusiveness in Bangla texts. Among the approaches, GPT-4o combined with CoT prompting consistently achieved the most reliable results. Moreover, to address inclusiveness, we introduced a curated dataset, named BanglaEduInclusive, comprising 3877 sentences, chunked into 338 paragraphs while preserving contextual integrity. The insights from this research work address a social problem and a potential pathway to solve it with the help of large language models.**

***Index Terms*—Inclusiveness, Textbooks, Large Language Model, Prompt Techniques**

I. Introduction

Inclusiveness is one of the most vital factors to ensure equality and fair representation in all aspects of life. In the manuscripts, textbooks, pamphlets even in the constitutions, the need to assess the representation of people from all gender, religions, culture, ethnicities and so on is now an essential issue, which is creating debates and conflicts between groups in the real world. In particular, inclusiveness in educational materials is very crucial for shaping diversity and equality during the early stage of a student's life. As the seventh most spoken language globally, an assessment of inclusiveness is also essential for all forms of communication in Bangla language, specially in the context of education.

To achieve fair representation in the field of education in Bangla, we must analyze the inclusivity of textbooks. Despite the importance of inclusiveness in education, we have not found any formal research work regarding the systematic detection of inclusiveness in this regard. There are different parameters to evaluate inclusiveness, such as gender, ethnicity, religion, culture, socioeconomic status, disability, etc. Manually assessing the texts using these parameters for inclusivity can be time-consuming, cognitively demanding, and possibly biased. In this context, Large Language Models (LLMs) offer a promising solution. Given their ability to process complex linguistic patterns and detect bias and representation across social groups, LLMs are well-suited to the task of assessing inclusiveness in Bangla texts. So we evaluated different LLMs with previous academic years Bangla textbooks to assess their ability to detect inclusiveness in the context. For this task we used 3 different state of the art models: GPT-4o (OpenAI), Gemini-1.5-Flash (Google AI), and Mistral Large 2.1 (Mistral AI).

Evaluating inclusiveness using LLMs is challenging due to its highly context-dependent nature, as inclusiveness can vary significantly across different situations, making it difficult to establish a consistent detection framework

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in Bangla. To address this challenge, instead of relying on computationally expensive fine-tuning methods, this study explores advanced prompting techniques as a cost-effective alternative to enhance the contextual understanding of LLMs. This approach offers a cost-effective solution without needing large amounts of additional resources. For this reason, we have experimented with three different prompting techniques to each LLM to maximize performance. The prompting techniques we used in our research are: Direct Prompting(DP), Prompting with Definition(PwD), and Chain-of-Thought (CoT).

We also introduce a novel labeled dataset 'BanglaEduInclusive' on Bangla inclusiveness detection, a task previously unexplored in the context of Bangla. The dataset is constructed from textbooks published by the National Curriculum and Textbook Board (NCTB), comprising the dataset which includes 3877 sentences, chunked into 338 paragraphs on 3 different parameters: gender, religion and culture.

Overall, the contributions of the paper are as follows:

1) We assessed the performance of different state-of-the-art LLMs in detecting inclusiveness in Bangla educational textbook materials.
2) We explored various prompting techniques to enhance contextual understanding in a cost-effective manner, without requiring additional resources.
3) We developed a dataset named *BanglaEduInclusive* from Bangla textbooks, labeled across different levels of inclusiveness based on three distinct parameters.

II. Background and Related Works

Different concepts of inclusion reflect varying logics regarding how their aspects are viewed [1]. In general, inclusiveness is defined as a process of organizing services and communities to respect differences, share resources and power equitably, and work toward empowerment and justice for all [2]. It works as a policy framework, but the interpretation and implementation of inclusion in practice remain challenging. For example, in the field of education, it requires a dual approach that focuses on both the rights of individuals, such as children, and the effectiveness of their experiences and education. This approach should shift beyond concerns about inputs and settings to a focus on outcomes, aiming to identify causal relationships [3].

There have been many works conducted in the field of inclusiveness detection. One study developed an algorithm to identify non-inclusive language in Spanish academic texts to promote gender-neutral writing, using a dictionary-based model to detect biased terms and train a classifier; however, it lacked deep contextual understanding [4]. Another study addressed this limitation by using LLMs for sexism detection on social media, combining fine-tuning with zero-shot prompting and employing multiple prompt-based models through a voting mechanism, which outperformed standalone fine-tuned transformers [5].

Other than that, in news articles, an open-source Python package was developed to detect bias by creating a dataset of biased texts and training a model for fairer analysis. This paper acknowledges a limitation that they have explored only a kind of bias rather than diversified areas [6]. Even in the context of software projects, GitHub Inclusifier helped developers identify non-inclusive language across several categories but its scope is limited to software artefacts and relies primarily on dictionary-based detection, without exploring the role of LLMs or prompting strategies for contextual inclusiveness detection [7].

So we wanted an approach that will be easier to implement, so that more people can use that technique to check inclusion in any textual data, only by providing prior context. Though scaling the size of a language models has shown to improve the range of benefits [8], it alone has not proved sufficient for achieving high performance on challenging tasks such as arithmetic, commonsense, and symbolic reasoning [9]. This is where comes prompt engineering, a challenging yet crucial task for optimizing the performance of large language models on customized tasks, which requires complex reasoning to examine the model's errors, hypothesize what is missing or misleading in the current prompt, and communicate the task with clarity [10]. There are a lot of prompt techniques available for better results. From a study [11], we came up with three prompt techniques: general prompting (which is also known as direct prompting), general prompts with definitions(PwD), and chain-of-thought (CoT) prompts [12].

Moreover, existing studies have examined bias in Bangla LLMs, with Sadhu et al. [13] focusing on gender and religious bias and Das et al. [14] introducing a dataset for cultural identity bias. While valuable, these works remain limited to bias detection in LLM outputs and do not address inclusiveness across broader social dimensions. Their datasets are also not oriented toward educational contexts, leaving a gap in understanding how inclusiveness is reflected in Bangla textbooks.

Large language models (LLMs) have achieved state-of-the-art performance across various NLP tasks, including question-answering, translation, and context-aware classification [8] [9]. Scaling models improves capabilities in comprehension and toxicity detection [9]. However, complex reasoning tasks such as arithmetic, commonsense, and symbolic reasoning remain challenging [10] [9]. Prompt engineering, including techniques like chain-of-thought prompting, has proven effective in guiding LLMs' reasoning and improving task-specific performance. LLMs have also been applied to context-sensitive tasks such as hate speech detection, where carefully designed prompts can leverage their knowledge to outperform conventional methods [11]. These advances highlight the potential of LLMs for analyzing nuanced properties in text, such as inclusiveness, though accessible and practical approaches

for this purpose are still limited.

Moreover, to the best of our knowledge, there has not been any work to assess the overall inclusiveness in the Bangla language.

III. Methodology

This chapter outlines the research methodology used to evaluate inclusiveness in NCTB textbooks using large language models (LLMs). The methodology is divided into 4 main sections: Data Collection, Data Annotation, Model Selection and Running and Prompting Techniques Study. Finally we evaluated the final outputs from the study, which is explained in another segment.



Fig. 1: Proposed Methodology for Data Curation

A. Data Collection from Textbooks

We had chosen multiple NCTB books for assessment of inclusiveness. We collected books that are more descriptive and contextual, rather than more factual or logical.

First, the sources were collected in PDF format from the NCTB website. Then Google Lens and i2OCR, OCR tools that convert scanned images and PDFs to editable text, were used to scrape the data manually with it. However, the use of OCR introduced significant challenges, as the extracted text often contained scattered characters, syntactic irregularities, and spelling errors. Consequently, a thorough manual formatting process was carried out to ensure textual accuracy and coherence. After the formatting process, the corpus was segmented into 338 distinct chunks. Those chunks were chosen in a manner that they have an independent context, so that every data point should hold a complete contextual meaning. This allows LLMs to evaluate inclusiveness based on the comprehensive context rather than isolated or fragmented portions of the text. In total, this process yielded 3877 sentences for analysis in the corpus.

B. Data Annotation

After formatting the data chunks, the process of data annotation was initiated. A crowd-sourced annotation approach was used for constructing the dataset. Though inclusiveness can be measured from multiple parameters, like gender, religion, economic class, cultural class, disabilities, regional perspective, this study primarily considered 3 broad parameters of inclusiveness: Gender, Religion and Culture. For each of the parameters, the textual data was annotated into one of these 5 predefined categories:

- Inclusive: Explicitly represents and includes all relevant communities under the given parameter based on the context.
- Marginal Inclusive: Does not explicitly mention all communities, but conveys a general sense or intent of inclusiveness.
- Non Inclusive: Clearly excludes or ignores certain communities under the parameter, while referencing only a few specific ones.
- Marginal Non-inclusive: Does not directly exclude any community, but the language or focus subtly implies the omission of some groups.
- Neutral: Sentences which are irrelevant or totally ambiguous in the context of the selected parameter.

This process was done by a group of 40 diverse people of annotators for better accuracy and performance in the detection of inclusivity in the selected chunks of the NCTB book excerpts. Each text segment was independently evaluated across all three parameters, resulting in a total of $3\times338=1{,}014$ annotations per annotator. To minimize subjectivity, we provided general annotation guidelines to ensure a consistent interpretation of the categories. Moreover, measurements for complete anonymity were taken so that any personal information of the annotators remained hidden and no annotator had visibility into others' labels. These measures were implemented to preserve the integrity and objectivity of the data labelling.

For the construction of the final dataset, annotations were aggregated using an unweighted majority voting technique, where all annotators' inputs were assigned equal weight. [15] Thus the final label for each data point was determined by the category receiving the highest number of votes. This process yielded the finalized labeled dataset of NCTB textbooks, which serves as the basis for subsequent analysis.

C. Model Selection and Running

We used 3 advanced and currently well-performing models like GPT-4o, Gemini-1.5-Flash, Mistral Large 2.1. The post-training alignment process of GPT-4o results in improved performance in measures of factuality [16]. Gemini 1.5 models are built to handle long contexts; they have the ability to recall and reason over fine-grained information [17]. Mistral Large 2.1 also performs better in reasoning than the previous models [18]. Therefore, based on the performance of reasoning, contextual understanding, and multilinguality, we selected these three models.

We used APIs of the state-of-the-art models for inference the LLMs, where the curated dataset, BanglaEduInclusive, was given as the input. The models then generated outputs, in the form of categorical labels for each data instance, showing their ability to understand, analyze, and classify the content based on the knowledge they had learned showing their internal reasoning and contextual representation.

Book Name	Edition	Chapters
Bangladesh and Global Studies Class 8	2018	5, 9, 10, 11
Bangladesh and Global Studies Class 8	2022	3
Bangladesh and Global Studies Class 9-10	2018	13, 14, 15
Physical Education and Health	2018	3
Physical Education and Health Class 9-10	2018	7
History and Social Science Class 7	2022	3, 4
History and Social Science Class 9-10	2022	7, 8

TABLE I: Textbooks and selected chapters used in the dataset.

D. Prompting Techniques

In this research work, our objective was to get the optimal performance with minimal resource in the measurement of inclusiveness using large language models. So we investigated various methodologies that could enhance the quality of outputs from the LLMs for this specific evaluation task. Though fine-tuning often provides substantial improvements in performance, it was impractical due to its significant computational demands and the necessity of large-scale training datasets [19]. As there is a scarcity of Bangla datasets, particularly those derived from educational materials related to finding inclusiveness, fine-tuning for our context was more constrained.

Given these limitations, we selected advanced prompting techniques that can be an alternative solution as they require neither additional datasets nor extensive computational resources [20]. Among the existing prompting strategies we implemented three methods: Direct Prompting [21], Prompting with Definition [11], and Chain of Thought (CoT) [12] to get better output generation.

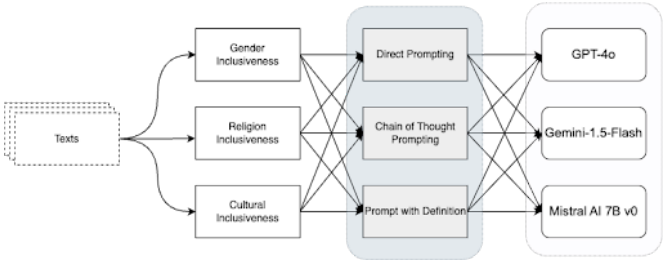


Fig. 2: Text processing with three parameters across three prompting techniques using three different LLMs.

Direct prompting, also called Zero-shot Prompting, is a technique that involves presenting the model with a clear and simple instruction for generating output [21]. It was used as the technique in general by most of the users as zero-shot closely aligns with natural human task execution [8]. The prompting with definition also uses a context with a question, but it augments the instruction with supplementary definitional information, enabling the LLM to to leverage additional context in its decision-making process [11]. CoT allows the model to decompose complex tasks into a series of intermediate reasoning steps, which means that additional computation can be allocated to problems that require more reasoning steps [12]. For each prompting technique, we designed tailored templates.

In Direct Prompting, we provided only the data instance along with a clear instruction to classify it into one of the five predefined categories. For Prompt with definition, we used both the data and conceptual definitions of inclusiveness dimensions and asked the LLMs to classify the data accordingly. Specifically, the definitions used were as follows:

Definition of Gender Inclusiveness: *Gender inclusiveness refers to the practice of creating an environment where individuals of all gender identities feel welcomed, respected, and valued. It is about creating a supportive and equitable environment that recognizes and values all gender identities, challenges stereotypes, and promotes acceptance and understanding among individuals* [22].

Definition of Religious Inclusiveness: *Religious inclusion refers to the practice of recognizing, respecting, and integrating diverse religious beliefs and practices within a society. It emphasizes the importance of allowing individuals from various religious backgrounds to participate fully in social, cultural, and political life without discrimination or exclusion* [23].

Definition of Cultural Inclusiveness: *To define cultural inclusiveness, it is recognized as a multidimensional concept influenced by various factors, including social norms, cultural identity, and environmental aspects. It reflects the degree to which individuals and groups from different backgrounds are respected and valued within a community or organization* [24].

Finally, for Chain of Thought prompting, we used a test case for labeling the data with proper contextual understanding steps. The CoT technique was designed in a way so that the LLMs can decompose the primary task into subproblems and approach the whole classification task in a step by step manner [12].

Upon execution, each model was evaluated across the dataset using three inclusiveness parameters and three distinct prompting strategies, ultimately producing nine separate outputs, which correspond to the combination of parameters and prompting techniques. When aggregated across all three models, this process resulted in a total of twenty-seven labels being generated for each data instance. These outputs were subsequently evaluated to assess the effectiveness of the prompting strategies and the overall performance of the models in measuring inclusiveness.

IV. Evaluation

The performance of the LLMs was evaluated by their generated outputs with the manually annotated dataset. To evaluate the LLMs we employed 4 widely used metrics: accuracy, precision, recall and F1 score.

1) Output Analysis of GPT-4o on Different Prompting Techniques: Assessing the data, we can clearly find that for the GPT-4o model the COT prompting technique gives the best prediction output as it has the best accuracy and F1 score compared to DP and PwD.

2) Output Analysis of Gemini-1.5-Flash on Different Prompting Techniques: Whereas for the Gemini-1.5-Fl model, the CoT prompting technique also produces the most effective prediction outputs, achieving the highest accuracy compared to DP and PwD. However, the performance analysis clearly indicates that Gemini struggles with classifying for the 'Culture' parameter but demonstrates comparatively stronger performance for 'Religion' and 'Gender'.

3) Output Analysis of Mistral Large 2.1 on Different Prompting Techniques: Unlike the other LLMs, Mistral displays a distinct trend. It performs better in detecting religious inclusiveness when using Prompting with Definition (PwD) compared to Direct Prompting (DP) or Chain of Thought (CoT). However, for the other two parameters, it still achieves the best performance with CoT, consistent with the behavior of the other models.

V. Findings

After evaluating the curated annotated data with three distinct prompt techniques on three different LLMs on three parameters, we wanted to come to a conclusion on which model performs best in the detection of inclusiveness in Bangla language.

For the gender inclusiveness parameter, all three models- GPT-4o, Gemini-1.5-Flash, and Mistral Large 2.1 demonstrate nearly identical performance when using Prompting with Definition (PwD) and Chain of Thought (CoT) techniques. However, Mistral Large 2.1 and Gemini-1.5-Flash exhibit lower performance when using Direct Prompting (DP) for detecting gender inclusiveness than GPT-4o. Regarding religious inclusiveness, all models achieve their highest performance across prompting techniques, with the exception of Mistral Large 2.1 using the CoT method. Specifically, Mistral achieves 69% accuracy in detecting religious inclusiveness with CoT, whereas GPT-4o and Gemini-1.5-Flash achieve 87% and 85%, respectively. For cultural inclusiveness, the models perform worse compared to the gender and religious parameters. Additionally, all models exhibit the lowest accuracy with the PwD method relative to DP and CoT. This trend highlights an area for improvement in future development and optimization of large language models.

Moreover, for each model, we calculated the average of accuracy as well as the F1 score for each prompt technique of all parameters. Here all the best values for each model are indicated as bold, and the best values for each prompting across all the models are indicated with an underline.

TABLE V: Overall accuracy across different prompting techniques and models.

Prompt-type	Model	Overall Accuracy
Direct Prompting	GPT	**0.683**
Direct Prompting	Gemini	0.580
Direct Prompting	Mistral	0.603
Prompting with Definition	GPT	**0.660**
Prompting with Definition	Gemini	0.603
Prompting with Definition	Mistral	0.627
Chain-of-Thought (CoT)	GPT	**0.763**
Chain-of-Thought (CoT)	Gemini	0.743
Chain-of-Thought (CoT)	Mistral	0.653

For each model, we can clearly see that the Chain-of-Thought (CoT) technique is outperforming every other method of prompting, indicating a clear gap in the fields of reasoning inclusiveness on Bangla text data. We can also observe that GPT-4o exhibits lower performance with Definition-Based Prompting (PwD) than Direct Prompting(DP), whereas Gemini and Mistral indicate an improvement from DP to PwD.

From these observations, we can say that, with proper reasoning and logical understanding to solve the problem of assessing inclusiveness in Bangla text data, the models can improve their accuracy by a significant amount. The accuracy rates for the CoT prompting in each model are the proof for this conclusion.

Moreover, to address the lower performance by PwD for the GPT-4o model, we come up with an observation that the definition for inclusiveness shortens the window of analysis of the inclusiveness in a context. If the definition does not provide all the angles for inclusiveness in a particular scenario, that can lead to a shortened scope of assessment, ultimately performing poorly for a particular situation.

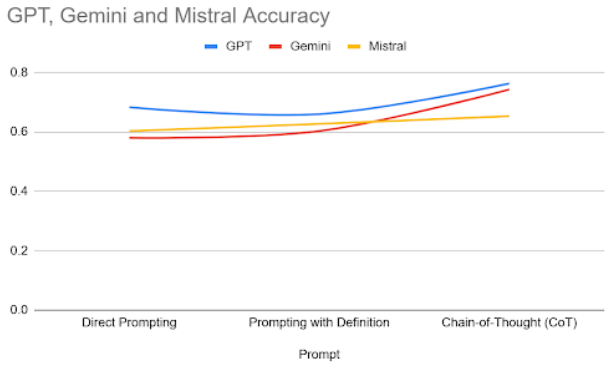


Fig. 3: Accuracy of GPT, Gemini, and Mistral across three prompting techniques.

To verify that our remarks and observations are correct and further specify the conclusions, we also assessed F1

TABLE II: Performance of GPT-4o on different parameters and prompt types

Model	Parameter	Prompt-type	Overall Accuracy	Precision	Recall	F1 Score
GPT-4o	Gender	DP	0.72	0.84	0.75	0.78
GPT-4o	Gender	PwD	0.66	0.80	0.69	0.68
GPT-4o	Gender	CoT	0.77	0.82	0.80	0.79
GPT-4o	Religion	DP	0.81	0.92	0.83	0.87
GPT-4o	Religion	PwD	0.87	0.90	0.90	0.90
GPT-4o	Religion	CoT	0.88	0.89	0.91	0.90
GPT-4o	Culture	DP	0.52	0.76	0.55	0.63
GPT-4o	Culture	PwD	0.45	0.74	0.47	0.51
GPT-4o	Culture	CoT	0.64	0.72	0.67	0.67

[a]Overall Accuracy, Precision, Recall, and F1 Score are reported per parameter and prompt type.

TABLE III: Performance of Gemini-1.5-Flash on different parameters and prompt types

Model	Parameter	Prompt-type	Overall Accuracy	Precision	Recall	F1 Score
Gemini-1.5-Flash	Gender	DP	0.41	0.90	0.43	0.58
Gemini-1.5-Flash	Gender	PwD	0.65	0.92	0.68	0.71
Gemini-1.5-Flash	Gender	CoT	0.74	0.87	0.77	0.76
Gemini-1.5-Flash	Religion	DP	0.80	0.93	0.83	0.87
Gemini-1.5-Flash	Religion	PwD	0.85	0.90	0.88	0.89
Gemini-1.5-Flash	Religion	CoT	0.87	0.87	0.90	0.89
Gemini-1.5-Flash	Culture	DP	0.53	0.77	0.56	0.64
Gemini-1.5-Flash	Culture	PwD	0.31	0.70	0.33	0.38
Gemini-1.5-Flash	Culture	CoT	0.62	0.40	0.65	0.50

[a]Overall Accuracy, Precision, Recall, and F1 Score are reported per parameter and prompt type.

TABLE IV: Performance of Mistral Large 2.1 on different parameters and prompt types

Model	Parameter	Prompt-type	Overall Accuracy	Precision	Recall	F1 Score
Mistral Large 2.1	Gender	DP	0.58	0.83	0.61	0.69
Mistral Large 2.1	Gender	PwD	0.68	0.82	0.71	0.69
Mistral Large 2.1	Gender	CoT	0.73	0.78	0.76	0.77
Mistral Large 2.1	Religion	DP	0.80	0.93	0.83	0.87
Mistral Large 2.1	Religion	PwD	0.84	0.89	0.87	0.87
Mistral Large 2.1	Religion	CoT	0.69	0.83	0.72	0.77
Mistral Large 2.1	Culture	DP	0.43	0.72	0.45	0.52
Mistral Large 2.1	Culture	PwD	0.36	0.71	0.38	0.44
Mistral Large 2.1	Culture	CoT	0.54	0.69	0.56	0.57

[a]Overall Accuracy, Precision, Recall, and F1 Score are reported per parameter and prompt type.

score for each model, in respect to each prompt technique. F1 score is a comprehensive assessment matric, which gives better evaluation even if the data is not well distributed.

F1 Score

TABLE VI: Overall F1-score across different prompting techniques and models.

Prompt-type	Model	Overall F1-Score
Direct Prompting	GPT	**0.760**
Direct Prompting	Gemini	0.697
Direct Prompting	Mistral	0.693
Prompting with Definition	GPT	**0.697**
Prompting with Definition	Gemini	0.660
Prompting with Definition	Mistral	0.670
Chain-of-Thought (CoT)	GPT	**0.787**
Chain-of-Thought (CoT)	Gemini	0.717
Chain-of-Thought (CoT)	Mistral	0.703

From this table, it is evident that the Chain-of-Thought (CoT) technique outperforms the other prompting techniques across all three models. The evaluation results for F1 score consistently show that CoT achieves higher scores compared to DP and PwD. The increments in the accuracy metrics are observed for GPT-4o (0.787), Gemini-1.5-Flash (0.717), and Mistral Large 2.1 (0.703), which validates our observation that the models can better understand the overall context when guided with proper reasoning and logical multi-step explanations. Therefore, we conclude that using GPT with the Chain-of-Thought (CoT) prompting technique yields the best performance in assessing the inclusiveness of Bangla texts.

Here, this graph shows the overall performance of the models based on each prompt parameter in the assessment of inclusiveness detection in NCTB textbooks. The circles inclined to more outward form are performing better. Here, we can see that the blue graph explicitly shows an outward tendency in almost all the prompting techniques indicating the better contextual understanding in the assessment of inclusiveness in NTCB book texts.

We can conclude our observation by stating that, using the GPT-4o model with the Chain-of-Thought (CoT) prompting technique, we can find the best output for assessment of inclusiveness in the NCTB textbooks.

VI. Limitations and Future Work

This study offers valuable insights into assessing inclusiveness in Bangla texts but has several limitations that

suggest directions for future research.

- **Data Scarcity:** The dataset was limited and mostly neutral or inclusive, making balanced evaluation challenging. Future work could curate larger, more diverse datasets, including textbooks from all Bangla medium classes, for a more comprehensive assessment.
- **Scope of Inclusiveness Assessment:** This study assessed inclusiveness in Bangla texts using only three parameters: Gender, Religion, and Culture, which do not fully represent all marginalized communities. Expanding to include Ethnicity, Disability, Age, Socioeconomic Status, Language, Geographical Diversity etc. would provide a more thorough evaluation.
- **Prompting Techniques:** Only three prompting strategies: Direct Prompting, Prompting with Definition, and Chain-of-Thought (CoT) were explored in this research. Future research could examine alternative approaches, such as few-shot or instruction-based prompting, to improve model performance and generalizability.

VII. Conclusion

In conclusion, this study represents the first extensive study of assessment of large language models (LLMs) for inclusiveness detection in Bangla language textbooks with various prompting techniques. It addresses a significant gap in the literature by providing a systematic evaluation of different prompting strategies and their relative effectiveness in this domain. Our findings indicate an overall improvement in inclusiveness detection across the evaluated LLMs, achieved without extensive computational resources. Specifically, GPT-4o emerged as the most effective model, while Chain-of-Thought (CoT) was identified as the most successful prompting technique. The insights gained from this study offer valuable guidance for enhancing the reliability and effectiveness of inclusiveness detection in educational materials, particularly as LLMs become increasingly integrated into language-based applications.

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